

1

Generate and insert 200 random rows in your airport database.

```
INSERT INTO airport (airline_code, name, country, state, city, created_at, update_at)
SELECT 'Z' || CHR(65 + (g / 26) % 26) || CHR(65 + g % 26), 'Airport ' || g,
  (ARRAY['Kazakhstan','Turkey','France','Poland','Brazil','United Kingdom'])[1+FLOOR(RANDOM()*6)::int],
  CASE WHEN RANDOM() < 0.5 THEN NULL ELSE 'State ' || (1+FLOOR(RANDOM()*20)::int) END,
  (ARRAY['Astana','London','Tokyo','Mlawe','Kepuh','Paris','Berlin'])[1+FLOOR(RANDOM()*7)::int],
  NOW(), NOW()
FROM GENERATE_SERIES(1, 200) AS g;
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'airport' table with columns: airport_id (integer, PK), airline_code (character varying (10)), name (character varying (100)), country (character varying (100)), created_at (timestamp without time zone), update_at (timestamp without time zone), state (character varying (100)), and city (character varying (100)).

The main pane shows a query window with the following SQL query:

```
SELECT * FROM airport ORDER BY airport_id DESC LIMIT 10;
```

The Data Output pane displays the results of the query, showing 10 rows of data. The status bar at the bottom indicates 'Total rows: 10' and 'Query complete 00:00:00.220'.

airport_id	airline_code	name	country	created_at	update_at	state	city
234	ZHS	Airport 200	France	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	[null]	London
233	ZHR	Airport 199	Kazakhstan	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	State 8	Tokyo
232	ZHQ	Airport 198	Brazil	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	[null]	Berlin
231	ZHP	Airport 197	Kazakhstan	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	[null]	Tokyo
230	ZHO	Airport 196	France	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	[null]	Berlin
229	ZHN	Airport 195	Poland	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	[null]	London
228	ZHM	Airport 194	Turkey	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	State 14	Astana
227	ZHL	Airport 193	France	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	State 16	Astana
226	ZHK	Airport 192	Brazil	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	[null]	Paris
225	ZHJ	Airport 191	Kazakhstan	2026-09-29 10:11:25.720005	2026-09-29 10:11:25.720005	[null]	Mlawe

2

Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.

```
INSERT INTO airline_2 (airline_code, name, country, created_at, update_at)
VALUES ('KAZ', 'KazAir', 'Kazakhstan', NOW(), NOW());
```

The screenshot shows the pgAdmin 4 interface with the following details:

- Object Explorer:** The left sidebar shows the database structure. Under 'Tables (9)', 'airline_2' is expanded, showing its columns: `airline_id` (integer), `airline_code` (character), `name` (character varying), `country` (character varying), `created_at` (timestamp), and `update_at` (timestamp). Below it, the 'airport' table is also expanded, showing its columns: `airport_id` (integer), `airline_code` (character), `name` (character varying), `country` (character varying), `created_at` (timestamp), and `update_at` (timestamp).
- Query Editor:** The central pane shows a SQL query: `SELECT * FROM airline_2 WHERE name = 'KazAir';`
- Data Output:** The bottom pane displays the query results in a table with 6 columns: `airline_id` [PK] integer, `airline_code` character varying (10), `name` character varying (100), `country` character varying (100), `created_at` timestamp without time zone, and `update_at` timestamp without time zone. The results show one row with the following values: 1, 3, KAZ, Kazakhstan, 2026-09-29 10:14:30.645429, and 2026-09-29 10:14:30.645429.
- Messages:** A green message bar at the bottom right states: 'Successfully run. Total query runtime: 150 msec. 1 rows affected.'
- Status Bar:** The bottom status bar indicates 'Total rows: 1', 'Query complete 00:00:00.150', and 'CRLF Ln 1, Col 47'.

3

Update the airline country "KazAir" to "Turkey".

```
UPDATE airline_2
SET country = 'Turkey', update_at = NOW()
WHERE name = 'KazAir';
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'airline_2' table. The main query editor shows the SQL command: `SELECT * FROM airline_2 WHERE name = 'KazAir';`. Below the query editor, the Data Output tab displays the results of the query in a table format.

airline_id	airline_code	name	country	created_at	update_at
3	KAZ	KazAir	Turkey	2026-09-29 10:14:30.645429	2026-09-29 10:15:30.98244

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.127'. The system tray at the bottom shows the time as 10:15:51 on 29.09.2026.

4

Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".

```
INSERT INTO airline_2 (airline_code, name, country, created_at, update_at)
VALUES ('AEZ', 'AirEasy', 'France', NOW(), NOW()),
      ('FHG', 'FlyHigh', 'Brazil', NOW(), NOW()),
      ('FFY', 'FlyFly', 'Poland', NOW(), NOW());
```

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, including the 'airline_2' table with columns: airline_id, airline_code, name, country, created_at, and update_at. The central pane shows the SQL query: `SELECT * FROM airline_2 ORDER BY airline_id DESC LIMIT 5;`. The bottom pane displays the query results in a table format.

airline_id	airline_code	name	country	created_at	update_at
1	6	FFY	FlyFly	Poland	2026-09-29 10:17:00.755604
2	5	FHG	FlyHigh	Brazil	2026-09-29 10:17:00.755604
3	4	AEZ	AirEasy	France	2026-09-29 10:17:00.755604
4	3	KAZ	KazAir	Turkey	2026-09-29 10:14:30.645429

A status bar at the bottom indicates: "Successfully run. Total query runtime: 146 msec. 4 rows affected."

5

Delete all flights whose arrival in 2024 year.

```
DELETE FROM flight
WHERE schadule_arrival_time >= '2024-01-01'
AND schadule_arrival_time < '2025-01-01';
```

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the Object Explorer with a tree view of the database structure. The main pane shows a SQL query editor with the following query:

```
1 SELECT COUNT(*) FROM flight
2 WHERE schadule_arrival_time >= '2024-01-01' AND schadule_arrival_time < '2025-01-01';
```

The query results are displayed in the Data Output pane, showing a single row with a count of 0.

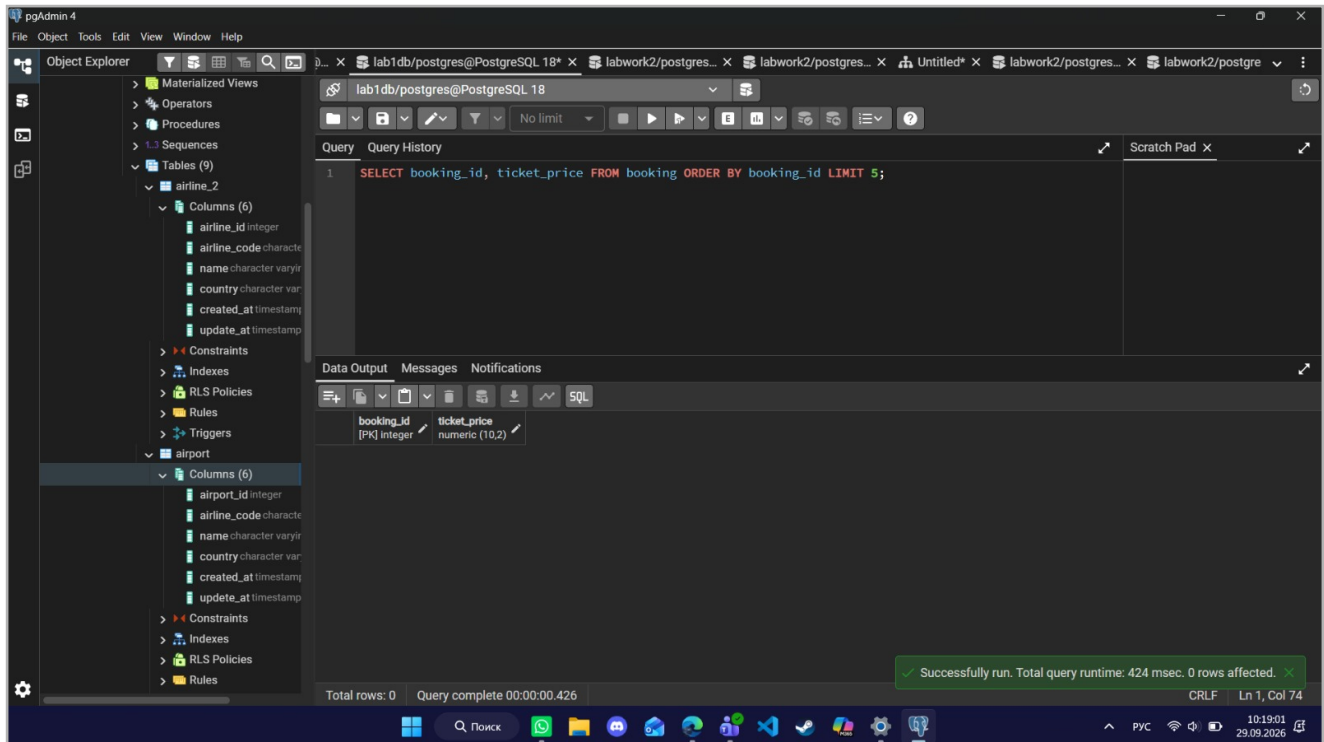
count
0

A status bar at the bottom indicates: "Total rows: 1 Query complete 00:00:00.150". A green message box at the bottom right states: "Successfully run. Total query runtime: 150 msec. 1 rows affected."

6

Increase the price of all tickets in booking table for flights by 15%.

```
UPDATE booking  
SET ticket_price = ROUND(ticket_price * 1.15, 2), updated_at = NOW();
```



7

Delete all tickets where price is less than 10000.

```
DELETE FROM booking
WHERE ticket_price < 10000;
```

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the 'Object Explorer' with a tree view of database objects. The 'airline_2' table is selected, showing its columns: airline_id (integer), airline_code (character), name (character varying), country (character varying), created_at (timestamp), and update_at (timestamp). The main query editor contains the following SQL query:

```
1 SELECT COUNT(*) FROM booking WHERE ticket_price < 10000;
```

The 'Data Output' pane at the bottom shows the results of the query:

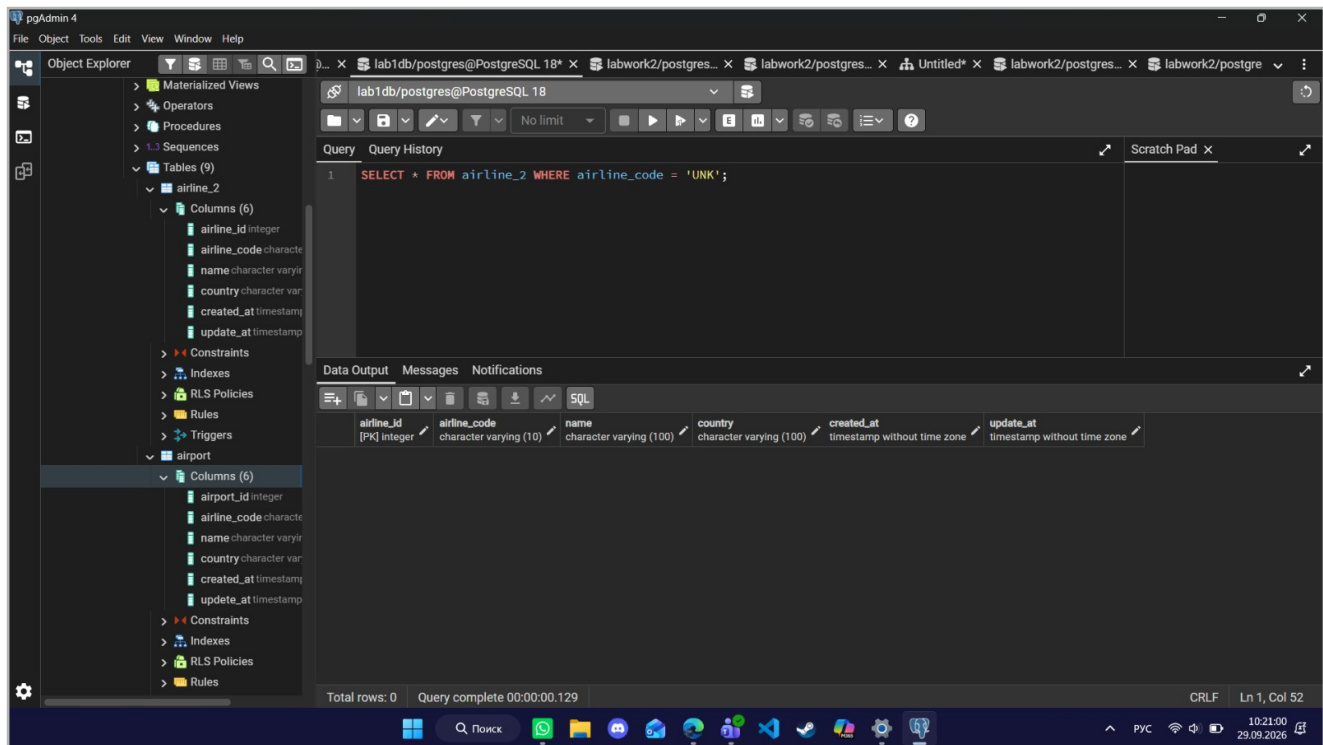
count
0

A green status bar at the bottom indicates: 'Successfully run. Total query runtime: 274 msec. 1 rows affected.' The bottom of the window shows the Windows taskbar with the time 10:19:52 on 29.09.2026.

8

Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).

```
UPDATE airline_2
SET airline_code = 'UNK', update_at = NOW()
WHERE airline_code IS NULL;
```



9

Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.

```
DELETE FROM baggage_check
WHERE created_at < '2023-06-01'
AND check_result = 'Not checked';
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including tables like 'airline_2' and 'airport'. The main query editor shows the following SQL query:

```
1 SELECT COUNT(*) FROM baggage_check
2 WHERE created_at < '2023-06-01' AND check_result = 'Not checked';
3
```

The Data Output pane at the bottom displays the results of the query:

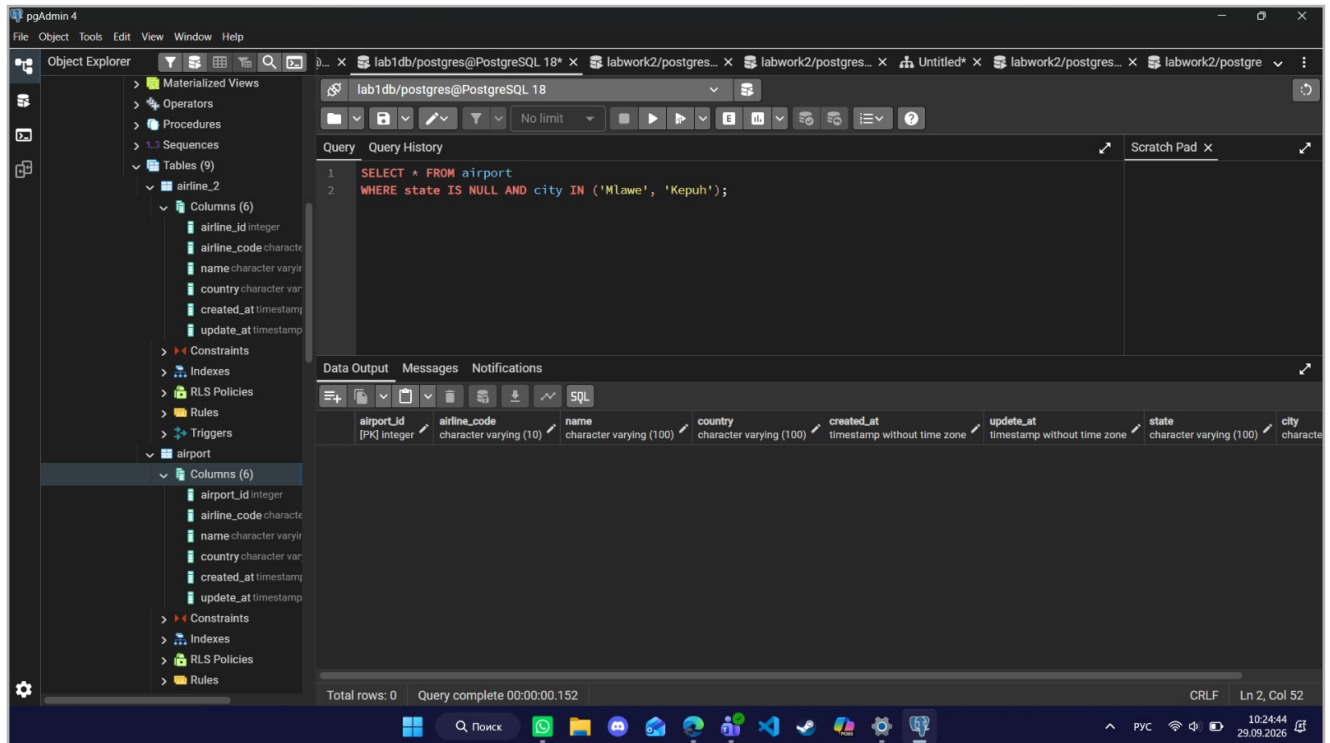
count
1

A status bar at the bottom indicates: "Total rows: 1 Query complete 00:00:00.167". A green message box at the bottom right states: "Successfully run. Total query runtime: 167 msec. 1 rows affected."

10

Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.

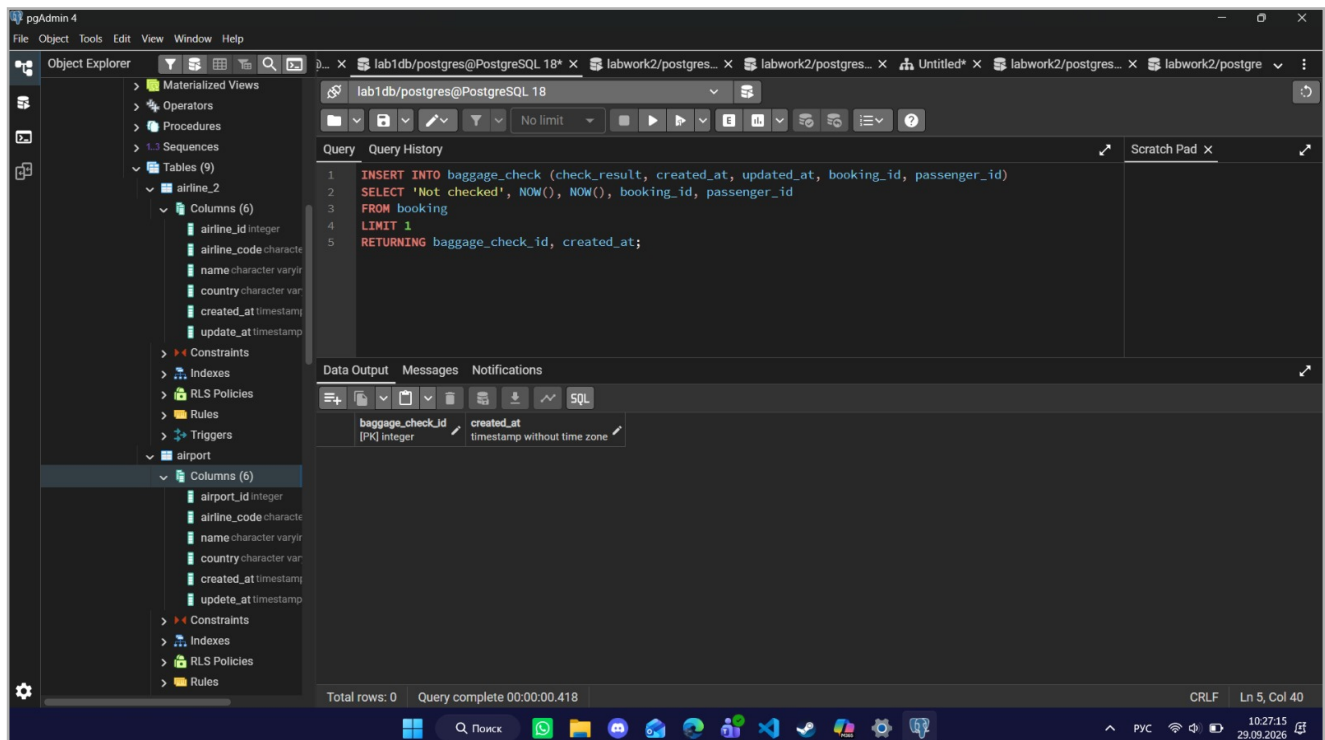
```
DELETE FROM airport
WHERE state IS NULL
AND city IN ('Mlawe', 'Kepuh');
```



11

Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.

```
INSERT INTO baggage_check (check_result, created_at, updated_at, booking_id, passenger_id)
SELECT 'Not checked', NOW(), NOW(), booking_id, passenger_id
FROM booking
LIMIT 1
RETURNING baggage_check_id, created_at;
```



12

Convert all country names in the airline table to uppercase to ensure data consistency.

```
UPDATE airline_2  
SET country = UPPER(country);
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including tables like 'airline_2' and 'airport'. The main query editor shows a SQL query: `SELECT airline_id, name, country FROM airline_2 ORDER BY airline_id;`. Below the query editor, the Data Output tab displays the results of the query in a table format. The table has three columns: 'airline_id' (integer), 'name' (character varying), and 'country' (character varying). The results show four rows of data.

airline_id	name	country
1	KazAir	TURKEY
2	AirEasy	FRANCE
3	FlyHigh	BRAZIL
4	FlyFly	POLAND

At the bottom of the interface, a status bar indicates: 'Total rows: 4', 'Query complete 00:00:00.165', and a green message box stating 'Successfully run. Total query runtime: 165 msec. 4 rows affected.'

13

Update both the name and country for the airline with `airline_id = 5`. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating `update_at` to the current date.

```
UPDATE airline_2
SET name = 'Global Airways', country = 'United Kingdom', update_at = NOW()
WHERE airline_id = 5;
```

The screenshot shows the pgAdmin 4 interface with the following details:

- Object Explorer:** The 'airline_2' table is selected under the 'lab1db' database. The table structure shows columns: `airline_id` (integer), `airline_code` (character), `name` (character varying), `country` (character varying), `created_at` (timestamp), and `update_at` (timestamp).
- Query Editor:** The query `SELECT * FROM airline_2 WHERE airline_id = 5;` is entered in the 'Query' tab.
- Data Output:** The query results are displayed in a table with the following columns: `airline_id`, `airline_code`, `name`, `country`, `created_at`, and `update_at`. The results show one row with the following values: `airline_id` is 5, `airline_code` is FHG, `name` is Global Airways, `country` is United Kingdom, `created_at` is 2026-09-29 10:17:00.755604, and `update_at` is 2026-09-29 10:28:41.321453.
- Messages:** A green message box at the bottom right states: 'Successfully run. Total query runtime: 113 msec. 1 rows affected.'
- Status Bar:** The bottom status bar shows 'Total rows: 1', 'Query complete 00:00:00.113', and the current position 'Ln 1, Col 46'.

14

Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'.

```
UPDATE airport
SET state = 'Capital District', update_at = NOW()
WHERE city IN ('Astana', 'London', 'Tokyo');
```

The screenshot shows the pgAdmin 4 interface with a query executed in the 'lab1db/postgres@PostgreSQL 18' database. The query is: `SELECT * FROM airport WHERE state = 'Capital District';`. The result is displayed in a table with 8 columns: `airport_id` (PK integer), `airline_code` (character varying (10)), `name` (character varying (100)), `country` (character varying (100)), `created_at` (timestamp without time zone), `update_at` (timestamp without time zone), `state` (character varying (100)), and `city` (character varying (100)). The table shows 14 rows of data, all with the state 'Capital District'.

airport_id	airline_code	name	country	created_at	update_at	state	city	
1	39	ZAF	Airport 5	Brazil	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Londr
2	41	ZAH	Airport 7	Turkey	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Londr
3	44	ZAK	Airport 10	France	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Astan
4	47	ZAN	Airport 13	Brazil	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Tokyo
5	48	ZAO	Airport 14	Brazil	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Tokyo
6	49	ZAP	Airport 15	France	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Tokyo
7	52	ZAS	Airport 18	Turkey	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Tokyo
8	54	ZAU	Airport 20	France	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Astan
9	59	ZAZ	Airport 25	Poland	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Tokyo
10	60	ZBA	Airport 26	Brazil	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Tokyo
11	64	ZBE	Airport 30	Turkey	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Astan
12	67	ZBH	Airport 33	Poland	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Londr
13	69	ZBJ	Airport 35	Brazil	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Londr
14	71	ZBL	Airport 37	Poland	2026-09-29 10:11:25.720005	2026-09-29 10:29:46.895646	Capital District	Astan

15

Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

```
UPDATE baggage_check
SET check_result = 'Checked', updated_at = NOW()
WHERE check_result = 'Not checked'
AND created_at >= '2024-03-01' AND created_at < '2024-04-01';
```

The screenshot shows the pgAdmin 4 interface with the following components:

- Object Explorer:** Displays the database structure, including tables like `airline_2` and `airport` with their respective columns and data types.
- Query Editor:** Contains the SQL query:

```
1 SELECT COUNT(*) FROM baggage_check
2 WHERE check_result = 'Checked'
3 AND created_at >= '2024-03-01' AND created_at < '2024-04-01';
4
```
- Data Output:** Shows the result of the query as a table with one row:

count
0
- Messages:** Displays a green message: "Successfully run. Total query runtime: 158 msec. 1 rows affected."
- Status Bar:** Shows "Total rows: 1" and "Query complete 00:00:00.158".